

Dina Abdel Rahman

Lead Data, AI & Cloud Architect

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Professional Summary

Experienced Data, AI and Cloud Architect with 19+ years of experience delivering complex data, analytics, AI and cloud solutions across public sector, energy, utilities, telecom and banking. Strong background in enterprise architecture, data platforms, cloud-native design, AI-enabled solutions, governance, security and large-scale transformation. Experienced in leading solution design from assessment to delivery, working with senior stakeholders, delivery teams and vendors across Agile and enterprise delivery environments.

Skills

Data Engineering, Analytics & AI Enablement:

- Delivered advanced analytics and AI-enabled use cases using Python, supporting machine learning (scikit-learn) and deep learning (PyTorch) frameworks.
- Designed data pipelines and analytical workflows that integrate batch, streaming and historical datasets, enabling predictive analytics, anomaly detection and operational intelligence.
- Applied data governance and quality frameworks to ensure trusted analytics, regulatory reporting readiness and auditability.
- Led AI solution development using Azure ML Studio and AI Foundry, integrating analytics capabilities into enterprise platforms.
- Supported model lifecycle enablement by designing deployment-ready analytics architectures aligned with enterprise security, operations and cost management constraints.

Architecture, Governance & Leadership:

- Acted as architecture lead and design authority on complex programmes, facilitating architecture reviews, option assessments and decision forums.
- Led architecture assessment and assurance activities, validating solution designs against enterprise principles, security standards and operational requirements.
- Advised senior stakeholders on technology evaluation and buy-vs-build decisions, balancing innovation with risk, cost and regulatory constraints.
- Supported Agile delivery, collaborating with development teams on backlog refinement, sprint planning, solution validation and release coordination.
- Produced clear, structured architecture documentation and executive-level reporting, enabling effective governance, knowledge transfer and long-term platform evolution.

Migration & Large-Scale Transformation

- Led and supported end-to-end data and platform transformation programmes, modernising legacy and fragmented data landscapes into scalable, cloud-native and hybrid architectures.
- Designed enterprise ingestion and integration architectures onboarding large numbers of heterogeneous data sources, spanning relational databases, NoSQL stores, files, APIs, logs and event streams.
- Architected and governed ETL/ELT pipelines and ingestion frameworks enabling reliable batch and near-real-time data processing at scale.
- Delivered migration-critical design activities including schema rationalisation, data normalisation, dependency analysis, and data quality enforcement, ensuring continuity of service and regulatory compliance.
- Defined backup, disaster-recovery and high-availability strategies aligned to SLA and operational resilience requirements for mission-critical data platforms.
- Produced target-state architectures and phased delivery roadmaps, balancing technical debt reduction, business value and delivery risk in complex enterprise environments.

Tools and Platforms:

- **Data Platforms:** Oracle, MySQL, PostgreSQL, Cassandra, Neo4J, Victoria Metrics, Redis, Elasticsearch, Teradata, Cloudera, Apache Spark, Apache Kafka, Apache Airflow, Microsoft Fabric, AWS Glue, AWS Athena, Alteryx
- **Programming & AI:** Python, FastAPI, Flask, scikit-learn, PyTorch, SageMaker, Azure ML Studio, AI Foundry, LangChain, LangGraph, CrewAI.
- **Analytics & Visualisation:** Power BI, MicroStrategy, Tableau
- **DevOps:** Git, Jenkins, GitLab, Azure DevOps, Terraform
- **Cloud Native:** Docker, Kubernetes, OpenShift

Experience:

Data & AI Solution Architect | KPMG | Ireland (May 2025 – Present):

- Deliver Data, AI and cloud solution architecture services across public sector, utilities, energy and banking engagements.
- Lead solution design across data architecture, data management, cloud modernisation, governance, security and AI-enabled platforms.
- Support clients across assessment, target-state architecture, solution options, delivery planning and architecture assurance.
- Work with business, data, IT, security and delivery teams to translate requirements into practical and scalable architecture designs.
- Contribute to tenders and business development activities, including solution design, delivery planning and resource estimation.

Main Projects

Enterprise AI Platform | Banking Client

- Delivered architecture design for an enterprise AI platform deployed on Azure.
- Defined the end-to-end platform architecture covering AI agents, orchestration, data integration, governance and operational tooling.
- Designed core platform modules, including agent runtime, knowledge ingestion, action execution and multi-agent coordination.
- Designed secure cloud infrastructure covering networking, data storage, model hosting, monitoring and environment separation.
- Embedded security, cost management and observability controls to support secure and scalable platform operations.

Sustainability Data & Technology Assessment | Energy Client

- Led a sustainability data and technology assessment for a large, regulated energy organisation.
- Reviewed the end-to-end data, technology and governance landscape supporting ESG and CSRD reporting.
- Assessed the existing sustainability reporting solution across performance, data quality, security, auditability and integration.
- Analysed ESG data flows across enterprise systems, analytics platforms, geospatial tools and spreadsheet-based processes.
- Led a buy-versus-build assessment and defined future-state architecture principles, data controls and ESG lifecycle requirements.

Automated Invoice Processing Architecture | Public Sector Client

- Designed Azure-based cloud architecture for automated invoice processing.
- Designed the end-to-end invoice intake and validation process using scanned invoice images.
- Used Azure Document Intelligence to support invoice classification and OCR-based data extraction.
- Defined the architecture for secure storage of extracted invoice data, processing artifacts and validation results.
- Designed security controls to protect invoice data, scanned documents and system-generated artifacts.

Core Data Platform Assessment | Public Sector Client

- Performed an end-to-end assessment of the client's core data system and supporting infrastructure.
- Reviewed database models, table structures, constraints, storage design, internal procedures and bulk load processing.

- Identified archive requirements and helped define the archive process and target archive solution.
- Assessed data quality issues across critical data elements and established data quality contract guidelines.
- Reviewed authentication, authorisation, security logging, monitoring, backup, high availability and disaster recovery arrangements.
- Recommended improvements to support service continuity, restoration, resilience and agreed SLA requirements.

Senior Solution Architect | Ericsson | Ireland (Nov 2022 – May 2025):

- Led the architecture design for the ESOA assurance sub-system, a next generation 5G network automation platform within the BSS landscape.
- Drove technical discussions with product, engineering and stakeholder teams to analyse requirements, risks, dependencies and solution options.
- Designed scalable cloud-native components, modular services and data flow diagrams for a distributed microservices-based environment.
- Assessed solution designs covering microservices, data stores, message bus integration, API specifications and platform dependencies.
- Designed and assessed data transformation flows, ETL logic, topic message schemas and API payload structures.
- Contributed to ML and Generative AI use cases, supporting solution design and implementation guidance for product efficiency improvements.
- Reviewed user management, monitoring, maintenance and operational requirements to support platform reliability and service assurance.
- Analysed integration points and aligned solution dependencies with internal teams and external counterparts.
- Produced technical solution documentation and reviewed designs with Product Managers, Product Owners and engineering teams.
- Supported development and testing teams through weekly planning, solution clarification and delivery alignment.

AI, Data Solution Architect | DU | UAE (May 2019 – Nov 2022)

- Led the architecture and implementation of DU's enterprise data lake, supporting large-scale ingestion, governance, analytics and AI use cases.
- Assessed and integrated 60+ data sources across RAN, core, transmission, IP, applications, probes, infrastructure and operational systems.
- Designed the ingestion architecture covering source connectivity, data structures, data frequency, normalisation, field extraction and data quality controls.
- Designed Kafka platform components, including topic structure, data schema, retention policy, compression and security controls.
- Designed Cloudera platform components, including HDFS structure, data formats, Spark, Hive, Impala jobs and Ranger security policies.
- Led AI and data project assessments covering use case identification, complexity analysis, data availability, ROI and delivery planning.
- Evaluated platforms for model training, inference and lifecycle management, supporting AI solution selection and implementation planning.
- Prepared POCs for AI, analytics and automation use cases, using stakeholder feedback to move concepts toward functional delivery.
- Managed engagement with business, technology, security, infrastructure, vendors, government entities and external partners.
- Led RFI/RFP activities for network assurance and automation platforms, including use case definition, vendor evaluation and scoring.
- Designed IoT and analytics solution proposals for telecom field operations, fleet management, industrial IoT and patrol management use cases.

Data Analytics and Engineering Lead | DU | UAE (August 2014 – May 2019)

- Led data analytics and engineering activities across network operations, performance management and operational reporting.
- Extracted, integrated and analysed high-volume data from probing systems, network platforms and operational systems.
- Developed SQL scripts, ETL logic and reporting datasets to support network investment, rollout planning and feature deployment decisions.

- Built real-time performance dashboards with low-latency updates to support operational monitoring and service assurance.
- Prepared network traffic forecasting and capacity planning analysis using device, coverage, technology and usage data.
- Analysed M2M and enterprise device performance to identify service trends, issues and optimisation opportunities.
- Represented network operations in discussions with business, marketing and senior stakeholders.
- Led a multi-year network upgrade programme across operations, coordinating 15+ stakeholders and 40+ team members across RAN, core, transport and infrastructure teams.

Migration Consultant | Huawei | Various Locations (November 2007 – July 2014)

- Delivered reporting and performance monitoring solutions across telecom network transformation programmes.
- Designed real-time dashboards and KPI reporting solutions to improve visibility of network performance and operational trends.
- Extracted, integrated and analysed high-volume network and operational data from multiple source systems.
- Developed structured reporting workflows to provide accurate performance metrics for operations teams and leadership.
- Analysed network performance data to identify trends, anomalies and optimisation opportunities.
- Supported cross-functional stakeholder coordination, ensuring consistent delivery of dashboards, reports and data insights.

Professional Training

- Kubernetes and Helm Administration
- Developing Cloud Native Applications
- Microsoft Azure DevOps Engineer
- Microsoft Azure AI Engineer
- Microsoft Azure Machine Learning Engineer
- Microsoft Azure Solution Architect Professional
- Microsoft Azure Data Engineering
- Developing on AWS Professional Training
- DevOps Engineering on AWS Professional Training
- AWS Machine Learning Engineer Professional Training
- Intel® Edge AI for IoT Developers

Education

BSc Electronics and Communication Engineering | Cairo University (2002 – 2007) General grade very good

Certifications

Microsoft Certified: Azure AI Engineer Associate | Microsoft (May 2026)

Microsoft Certified: Azure Data Fundamentals | Microsoft (March 2022)

Microsoft Certified: Azure AI Fundamentals | Microsoft (March 2022)

Microsoft Certified: Azure Fundamentals | Microsoft (Feb 2022)

AWS Certified Machine Learning – Specialty | Amazon Web Services (Dec 2021)

AWS Certified Solution Architect Associate | Amazon Web Services (Oct 2021)

AWS Partner: Accreditation (Business) | Amazon Web Services (Dec 2021)

AWS Partner: Accreditation (Technical) | Amazon Web Services (Dec 2021)

AWS Partner: Cloud Economics Accreditation | Amazon Web Services (Dec 2021)